Section A

1 **(a)** $CH_3(CH_2)_6CH_3$ (labeled highest b.p.)
$(CH_3)_3C\text{-}C(CH_3)_3$ (labeled lowest b.p.)
& 1 other structure
Reasons:
$CH_3(CH_2)_6CH_3$ has the longest carbon chain with greatest surface area for van der Waals' forces to operate
$(CH_3)_3C\text{-}C(CH_3)_3$ is the most branched with the smallest surface area

(b) **(i)** C_8H_{18} + 25/2 O_2 à 8CO_2 + 9H_2O
114 g C_8H_{18} requires 12.5 x 24 dm^3 O_2 = 300 dm^3 O_2
57 g C_8H_{18} requires 150 dm^3 O_2
≡ 150 x 100/20 dm^3 air = 750 dm^3 air

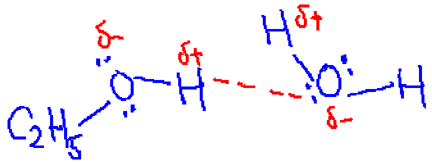
(ii) Octane undergoes substitution with bromine.
Brown bromine turns colourless & misty fumes (of HBr)

(c) **(i)** higher
Dipole-dipole attraction btw RCl molecules stronger than van der Waals' forces btw ethane molecules

(ii) higher
Any isomer of C_8H_{18} will be larger with more electrons.
Stronger van der Waals' forces

(6 max 5)

(d)



Show appropriately d+ and d- for O—H in each molecule & lone e pairs on O atoms
Show & label attraction btw H and lone pair of 2nd O atom

(e) **(i)** *s -bond:* single covalent bond formed by end-on overlap between s & s, s & p, p & p orbtals / diagrams
p-bond : covalent bond formed by sideways overlap between p-orbitals / diagram

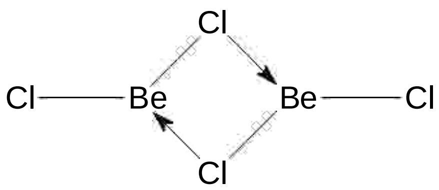
(ii) 5 σ and 1 π bond

Section B

2 **(a)** **(i)**

(ii) Be in $BeCl_2$ is 4 electrons short of an octet (or is electron deficient)..
It accepts electron pairs from Cl of another $BeCl_2$ molecule by dative bonding

(iii)



$\angle$Cl-Be-Cl = 120^o
$\angle$Be-Cl-Be = 105^o (or 104.5^o)

(b) **(i)** Energy is needed to <u>overcome attractive forces between the electrons and the protons</u> in the nucleus/molecule.

(ii) -ve plate is where $Be_2Cl_4^+$ is deflected towards; other plate is +ve

(iii) Draw correct pathway with label.
Deflection towards -ve plate because $BeCl_2^+$ is positive ion.
Deflection larger than for $Be_2Cl_4^+$ because $BeCl_2^+$ is lighter.

3 **(a)** For the reaction aA + bB à products

Rate equation is rate = $k[A]^x[B]^y$
x and y are the orders of reaction with respect to A and B respectively
k is the rate constant

(b) **(i)** ester

Comparing experiments 1 & 2, when [ester] doubles, rate also doubles.
$\Rightarrow$ reaction is <u>first order</u> with respect to ester.

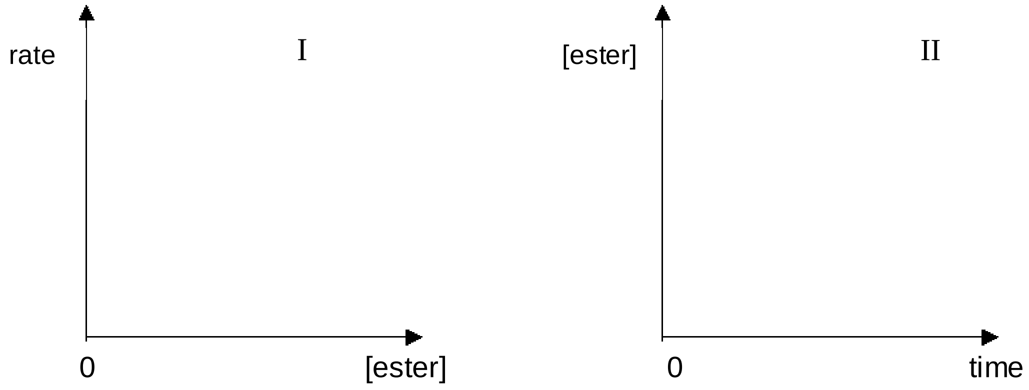
bromine
Comparing experiments 1 & 4, when [Br_2] doubles, rate increases 4 times.
$\Rightarrow$ reaction is <u>second order</u> with respect to Br_2.

(ii) 5.86×10^{-3}

(iii) rate = k [ester] $[Br_2]^2$

(iv) Rate increases by <u>8 times</u>

(v)



(c) Rate of reaction increases.
At higher temperature, more molecules possess energy equal to or exceeding the activation energy

4 **(a)** **(i)** $2Na_2O_2$ + $2CO_2$ à O_2 + $2Na_2CO_3$

(ii) Volume of CO_2 produced by 6 submariners per day = 6 x 500 dm^3
= 3000 dm^3
Mole of CO_2 produced = 3000/24 dm^3 = 125 mol
Mole of Na_2O_2 required = 125 mol
Mass of Na_2O_2 required per day = 125 x 78 g
= 9750 g (or 9.75 kg)

(b) **(i)** CO_2 + 2NaOH à Na_2CO_3 + H_2O

(ii) Mole of CO_2 dissolved = 1.54/44.0 = 0.035 mol
= mole of Na_2CO_3 formed
Conc of Na_2CO_3 in the solution = 0.035 x (1000/50.0)
= 0.70 mol dm^{-3}

Section C

1	C	11	D	21	C
2	B	12	B	22	B
3	D	13	D	23	D
4	B	14	C	24	C
5	B	15	C	25	A
6	C	16	A	26	D
7	A	17	D	27	C
8	B	18	A	28	D
9	B	19	D	29	B
10	D	20	B	30	C